

two-year horizon by buying a three-month T-bill. In three months, the T-bill expires, but then we invest the proceeds in another three-month T-bill. If we *roll over* three-month T-bills over seven successive quarters, we will have reached the same horizon as directly investing in a two-year T-note. The two strategies are not identical—we are subject to roll-over risk when we invest in a series of T-bills, whereas we know the yield today on the two-year Treasury note. Nevertheless, the price of a three-month T-bill cannot move too far from the price of a two-year Treasury note. If the roll-over strategy is very cheap compared to the two-year Treasury note, arbitrageurs will buy T-bills and sell notes. Conversely, if T-bills are too expensive compared to notes, investors will sell T-bills and buy notes. Following this logic, the price of any long-term bond must be related to the price of any short-term bond.

Risk Premiums

We can decompose the movements of long-term yields into pure “expectations” components and components that embed risk premiums. According to the *Expectations Hypothesis* originally developed by one of the great economists of the early twentieth century, Irving Fisher (1896),

$$\text{Nominal long yield} = \underbrace{\text{average of expected short rates}}_{\text{Expectations Hypothesis}} + \text{risk premiums.} \quad (9.2)$$

If we interpret the three-month T-bill as the short rate, then the two-year Treasury note yield should be the average of eight three-month T-bill yields: the current three-month T-bill, the future three-month T-bill yield that will prevail in three months time, the future three-month T-bill yield that will start in six months time, and so on, up to the end of two years. This is the Expectations Hypothesis component. In the real world, this does not hold. One three-month T-bill today and seven future, roll-over three-month T-bill positions do not, on average, equal the two-year Treasury yield.¹⁷ There is risk in the rolled-over T-bill positions: the macro-economy may change and inflation may spike, volatility may change, or investors may become suddenly more risk averse. Not surprisingly, the risk premium components, especially in long-term Treasuries, can be quite large.

Another decomposition is based on the *Fisher Hypothesis*, also by Fisher (1930):

$$\text{Nominal long yield} = \underbrace{\text{real yield} + \text{expected inflation}}_{\text{Fisher Hypothesis}} + \text{risk premiums} \quad (9.3)$$

The pure Fisher Hypothesis states that the long yield is equal to the real yield and expected inflation and ignores the risk premium components. Like

¹⁷There is a very large literature testing the Expectations Hypothesis beginning with Fama and Bliss (1987) and rejecting it overwhelmingly for the United States.